General Linear Process - Convergence in Mean Square

Let $X_{tn} = \sum_{j=0}^{n} \alpha_j Z_{t-j}$

If X_{tn} converges in mean square to X_t then $\mathbb{E}[(X_{tn} - X_t)^2] \to 0$ as $n \to \infty$.

Consider

$$X_{tn} - X_t = \sum_{j=0}^{n} \alpha_j Z_{t-j} - \sum_{j=0}^{\infty} \alpha_j Z_{t-j}$$

$$= - \sum_{j=n+1}^{\infty} \alpha_j Z_{t-j}$$

Hence

$$\mathbb{E}[(X_{tn} - X_t)^2] = \mathbb{E}\left[\left(- \sum_{j=n+1}^{\infty} \alpha_j Z_{t-j}\right)^2\right]$$

$$= \sum_{j=n+1}^{\infty} \alpha_j^2 \, \mathbb{E}(Z_{t-j}^2)$$

because $\{Z_t\}$ is a white noise process

$$= \sigma_Z^2 \sum_{j=n+1}^{\infty} \alpha_j^2$$

Now we must assess whether or not

$$\sum_{j=n+1}^{\infty} \alpha_j^2 \to 0 \quad \text{as} \quad n \to \infty$$

A sufficient conditional for $\sum_{j=n+1}^{\infty} \alpha_j^2 \to 0$ as $n \to \infty$ is that $\sum_{j=0}^{\infty} \alpha_j^2 < \infty$.

(i.e. $\sum_{j=0}^{\infty} \alpha_j^2$ converges to some limit L)

$\therefore$ If $\sum_{j=0}^{\infty} \alpha_j^2 < \infty$ then $\sum_{j=n+1}^{\infty} \alpha_j^2 \to 0$ as $n \to \infty$.

Therefore if $\sum_{j=0}^{\infty} \alpha_j^2 < \infty$ then

$$\mathbb{E}\left[(X_{tn} - X_t)^2 \right] \to 0 \quad \text{as} \quad n \to \infty$$

i.e. X_{tn} converges in mean square to X_t.